EAGLYS

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Grant Proposal

00 Title

Encryption-Aware Quantization for Privacy-Preserving Signal Processing

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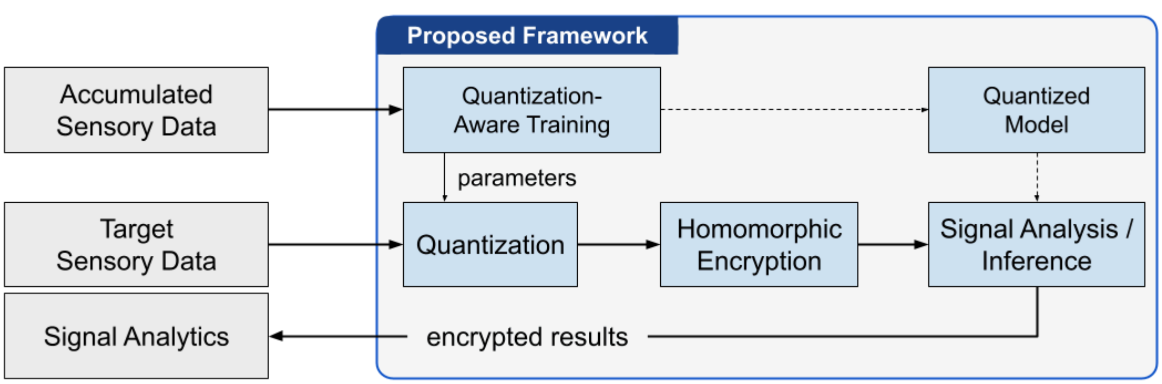
01 Executive Summary / Abstract

This research proposes an **encryption-aware quantization framework** for efficient privacy-preserving signal processing tasks. The goal is to investigate and design quantization schemes aligned with the numerical properties and noise constraints of the **CKKS homomorphic encryption** scheme. By utilizing a carefully selected combination of quantization and training techniques, the framework aims to reduce computational cost and ciphertext size while maintaining signal fidelity, and the performance of downstream tasks. The developed techniques are tuned to and **evaluated against CNN-oriented signal processing tasks**, such as acoustic environment classification and anomaly detection. A key benefit of this work is making encrypted processing of sensitive *time-series data* more efficient and deployable in real-world applications.



03 Background

Homomorphic Encryption

Homomorphic Encryption (HE) is a cryptographic technique that allows computations to be performed directly on encrypted data without decrypting it—enabling privacy preserving computations.

CKKS

Among the various homomorphic encryption schemes, CKKS—named after Cheon, Kim, Kim, and Song—is the most prominent approach for machine learning purposes due to its computational efficiency and support for vectorized operations. CKKS encodes real-valued data as scaled fixed-point numbers before encryption. While strict integer quantization is not required, limiting numerical precision through quantization-like techniques is practically necessary to manage the noise budget and computational depth of homomorphic operations.

Quantization

Quantization is a widely used technique in signal processing and machine learning for reducing computation, memory and energy costs while preserving model accuracy. Although these optimization goals are shared, quantization techniques and quantized models from machine learning cannot be directly applied as plug-and-play components in CKKS-based homomorphic processing pipelines. Instead, quantization must be carefully designed to account not only for quantization noise, but also for encryption-induced approximation error and precision losses arising from rescaling, bootstrapping, and noise accumulation across multiplicative depth. This motivates the design of encryption-aware quantization schemes that jointly optimize for signal fidelity, HE noise tolerance, and computational feasibility under encryption.

Signal Processing

Existing work on convolutional neural networks (CNNs) under homomorphic encryption has focused almost exclusively on image classification benchmarks and large language model acceleration, leaving signal processing — a domain with equally compelling privacy requirements and greater architectural compatibility with FHE — largely unexplored. Signal processing tasks, particularly acoustic classification, operate on real-valued, bounded, and structured data that aligns naturally with the fixed-point arithmetic of CKKS. Together, signal processing and CNNs define a problem space that is simultaneously practical — with clear privacy motivations in sensitive domains such as underwater acoustics — and technically tractable, offering a principled and feasible entry point for encrypted neural inference that the existing literature has yet to explore.

04 Research Objectives

RO1 Develop a principled **quantization scheme** that jointly accounts for quantization noise, CKKS approximation error, and noise accumulation across homomorphic operations, specifically tailored to CNN-based signal processing.

RO2 Design and train a 1D CNN architecture for acoustic classification and anomaly detection that is natively compatible with CKKS homomorphic encryption.

RO3 Identify and engage an industry or defense partner to co-define operational requirements and **validate** the developed pipeline **through a proof-of-concept** project, demonstrating practical viability with respect to task accuracy, inference latency, and cryptographic security level.

05 Project Structure

Year 1 - Foundations

- literature review: CNN + FHE + quantization; acoustic environment detection, etc.
- create cleartext baseline:
  - perform 1D CNN experiments, e.g., acoustic classification, speaker identification
  - dataset creation for performance measures
- create ciphertext baseline:
  - implementing CKKS-based pipeline accommodating multiple FHE engines: *Concrete ML* (Zama), *SEAL* (Microsoft) + in-house CKKS library (Eaglys)
  - measure and analyze out-of-the box performance
- contact potential industry partners, define PoC requirements

Year 2 - Core Contribution

- develop CKKS-aligned quantization-aware training (QAT) procedure
  - design encryption-aware quantization scheme
  - polynomial activation design for acoustic signals
- build internal prototype
  - validate technical choice against customer requirements
  - present pre-PoC demonstrator to partner
- first publication

Year 3 - System Integration

- apply CKKS pipeline to real-world tasks
  - delivery PoC demonstration
- integrate CKKS pipeline into a in-house Federate Learning (FL) framework
  - extend pipeline to encrypt model updates during FL aggregation
  - design secure aggregation protocol
- second publication

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Team Expertise and Feasibility

The feasibility of the proposed work is supported by the complementary expertise of our cryptography, artificial intelligence, and AI platform teams.

Our dedicated **cryptography team** with expertise in homomorphic encryption schemes, including TFHE, CKKS, and their practical implementation maintains an active research profile, with regular publications in peer-reviewed venues and participation in leading international conferences.

Our dedicated **AI team** brings expertise in machine learning model design, training, and optimization, with specific experience in image and acoustic signal analysis. The team also has a strong track record of working with industry partners, translating research outputs into practical demonstrations and proof-of-concept deployments.

The project further benefits from an established collaboration with an **industry partner** specializing in **hardware acceleration** for homomorphic encryption, providing access to optimized computational infrastructure that directly addresses the latency challenges inherent in FHE-based inference. Our platform team is uniquely positioned to bridge the gap between research-level cryptographic implementations and deployable real-world systems.

Lastly, the company brings direct experience from active federated learning projects, providing both the technical infrastructure and operational knowledge required for the privacy-preserving distributed inference objectives of this project

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Budget

Personnel The project appoints one full-time researcher dedicated to technical development and PoC delivery. This researcher is supported by a multidisciplinary team of senior researchers and PhD students who contribute on an in-kind basis, funded through existing funds.

Equipment A dedicated high-performance computing infrastructure is essential, as homomorphic encryption operations are significantly more computationally demanding than standard machine learning tasks.

Partner Engagement A dedicated budget is allocated for partner engagement activities, including travel to potential partner sites, meetings, and demonstration events.

	FTE	Year 1	Year 2	Year 3	Total	
Researcher	100%	60,000	60,000	60,000	180,000	USD
Principal Investigator	20%	15,000	15,000	15,000	45,000	USD
Equipment		10,000	2,500	2,500	15,000	USD
Partner Engagement		5,000	5,000	5,000	15,000	USD
Conference/Dissemination		0	5,000	5,000	10,000	USD
Total		90,000	87,500	87,500	**265,000**	USD

08 References

TODO